DO NOW — RECALL

Graphs and graph representations

Name: _____ Class: _____ Date: _____

1. Where is a value 8 inserted if BST root 10 has no left child?

2. What is the purpose of a next pointer in a linked list?

3. What does zero-based indexing mean?

4. In a BST root 20, left 10, right 30, where does 25 go?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

As the left child of 10.

2. Model answer

It identifies the next node.

3. Model answer

The first element is at index 0.

4. Model answer

Left child of 30.

Vocabulary — know these terms

vertex — A node in a graph.

edge — Connection between vertices.

adjacency list — Representation listing neighbours of each vertex.

adjacency matrix — Two-dimensional matrix marking which vertices are connected.

sparse graph — A graph with relatively few edges compared with possible connections.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Confirm and explain

Full lists: A:B; B:A,C; C:B,D; D:C.

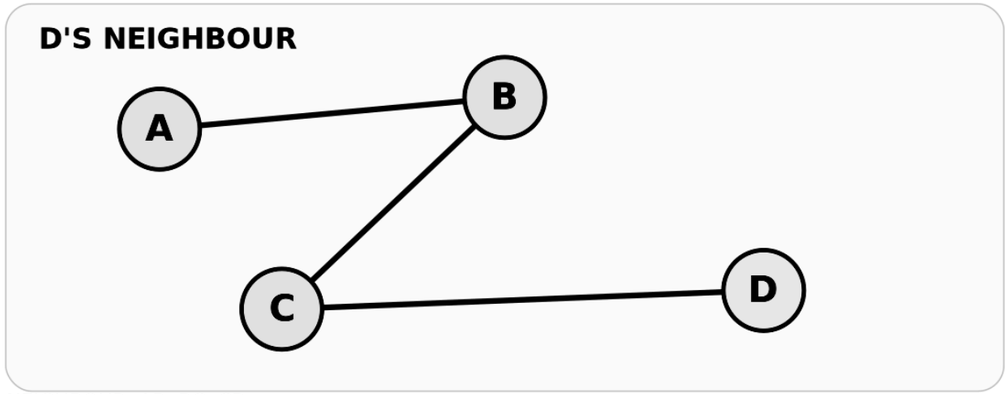
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Build adjacency lists for chain A–B–C–D.

A is adjacent to B; D is adjacent to C.



D'S NEIGHBOUR. AB, BC, CD

NEW LEARNING

Read, annotate and explain

Identify vertices

A, B, C and D are graph vertices.

Follow edges

Edges connect A–B, A–C and B–D. Because this graph is undirected, each connection works both ways.

Write adjacency lists

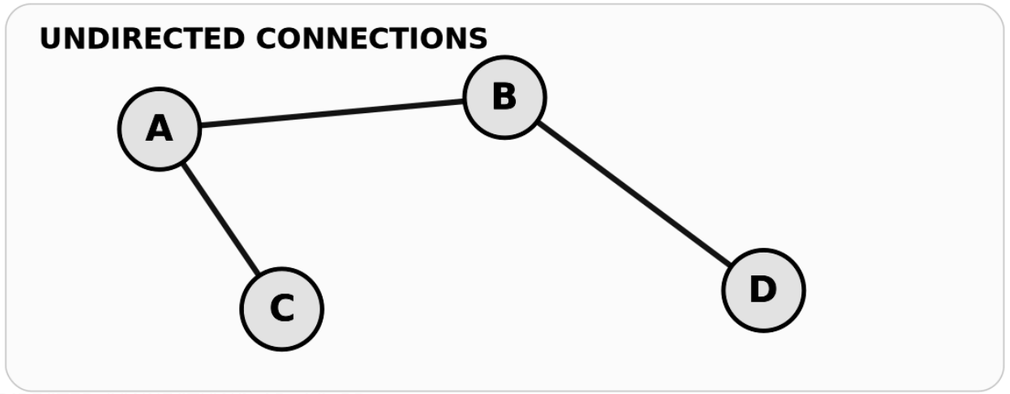
A:B,C; B:A,D; C:A; D:B. Every undirected edge appears in both endpoint lists.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

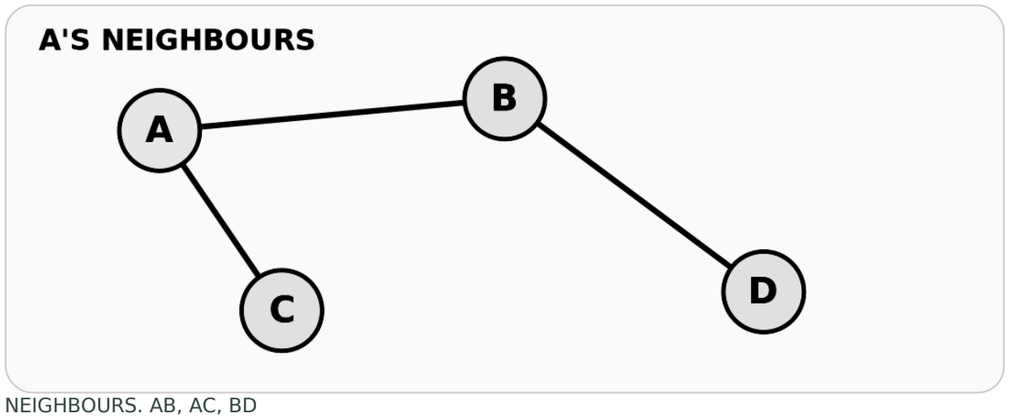
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



UNDIRECTED CONNECTIONS. AB, AC, BD

New Learning



A'S NEIGHBOURS. AB, AC, BD

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In an undirected graph, A–B is represented in both A's and B's adjacency lists.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

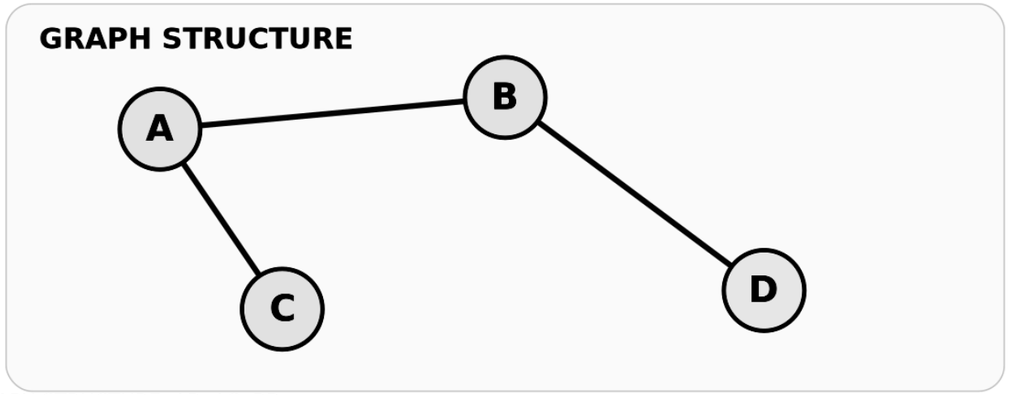
Question 3

Compare adjacency matrix and adjacency list storage on a sparse graph with many vertices.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



GRAPH STRUCTURE. AB, AC, BD

I DO — WORKED EXAMPLE

Follow the worked method

Build adjacency lists for triangle A–B, B–C, A–C.

Predict

Every vertex connects to the other two vertices.

Confirm and explain

A:B,C; B:A,C; C:A,B. Each undirected edge is listed from both endpoints.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name the vertices and number of edges in A–B, A–C, C–D.

Question 2

Give adjacency lists for an undirected graph with edges AB, BC, CD.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict A is adjacent to B; D is adjacent to C.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

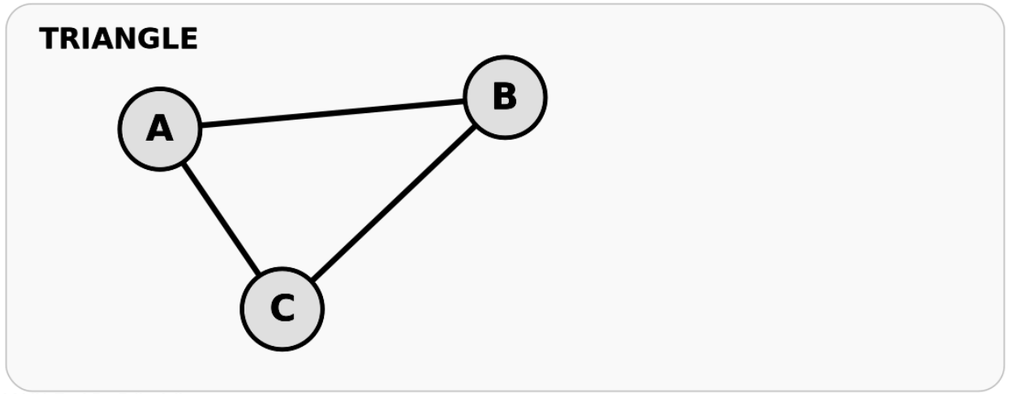
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Build adjacency lists for triangle A–B, B–C, A–C.

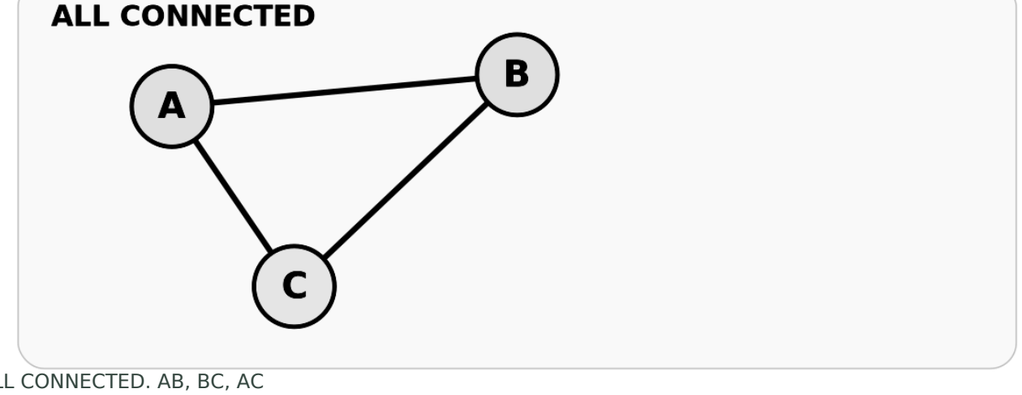
Every vertex connects to the other two vertices.



TRIANGLE. AB, BC, AC

Build adjacency lists for triangle A–B, B–C, A–C.

Every vertex connects to the other two vertices.



ALL CONNECTED. AB, BC, AC

WE DO — GUIDED PRACTICE

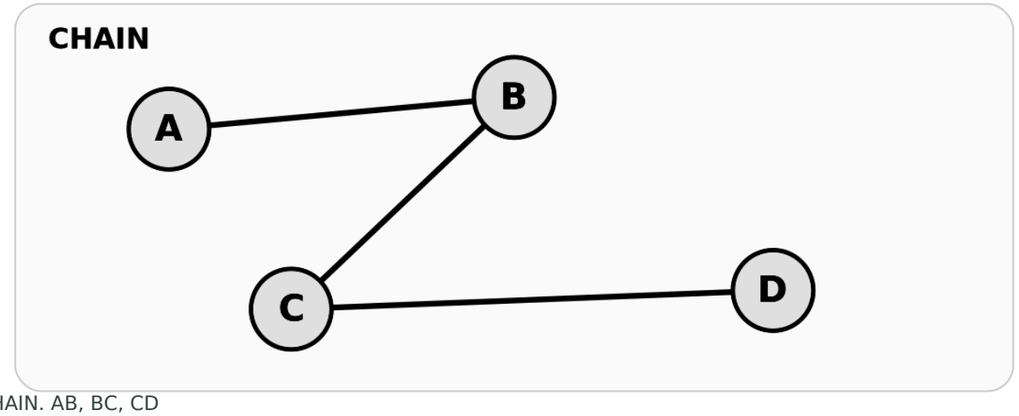
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Build adjacency lists for chain A–B–C–D.

A is adjacent to B; D is adjacent to C.



CHAIN. AB, BC, CD